

Final Progress Report

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GitHub: <https://github.com/SandeshJIT/rl-maze-solver>

1. What have you already achieved?

Since the last report, I have successfully implemented a **Deep Q-Network (DQN)** agent to address the limitations observed with Q-Learning, particularly catastrophic forgetting and poor generalization across multiple mazes. The DQN implementation uses a **Double DQN with a Dueling Network architecture**, where the model separates state value and action advantages to improve stability and reduce overestimation bias. I also implemented an efficient **Replay Buffer using a pre-allocated numpy array**, enabling fast sampling and better training performance. The state is encoded as a **flattened normalized maze grid combined with the agent's position**, allowing the model to generalize across different maze layouts without relying on a Q-table.

I trained the agent in both:

- **Single-maze setup** (fixed maze)
- **Multi-maze setup** (multiple mazes with different seeds)

During multi-maze training, I observed catastrophic forgetting when training too long on one maze. I addressed this by **frequently rotating mazes (every 100 episodes)**, which significantly improved generalization across all mazes.

I also extended the evaluation pipeline with metrics such as:

- Average reward
- Steps to reach the goal
- Success rate
- Loss and epsilon decay

Additionally, I created **visualizations (GIFs)** showing the trained agent solving mazes and ensured the DQN integrates seamlessly with the existing environment and visualization pipeline.

2. What are your immediate next steps?

My next step is to implement **Proximal Policy Optimization (PPO)**, a policy-gradient method that is better suited for handling sparse rewards and more complex exploration scenarios.

I plan to:

- Build a PPO agent compatible with the current environment
- Compare PPO performance against Q-Learning and DQN
- Analyze differences in convergence speed, stability, and generalization

3. Are there any challenges you are facing or adjustments you need to make?

A key challenge I encountered was **catastrophic forgetting in multi-maze training**, even with DQN. This required adjusting the training strategy (frequent maze rotation), which improved results significantly. Another consideration is **balancing exploration vs. exploitation**, especially in larger or more complex mazes. This may require further tuning of reward structures or training schedules.

No major changes are required to the original project scope, but I am focusing more on **generalization across environments** as a core evaluation metric.

4. What is your overall plan to complete this project?

Over the next two weeks, I plan to:

- Implement and train a **PPO agent**
- Run experiments comparing **Q-Learning, DQN, and PPO** across multiple mazes
- Analyze performance using consistent metrics (reward, steps, success rate)
- Build a **final visualization (multi-agent comparison or "race")** showing how different agents solve the same maze

The final submission will include a complete comparison of all three approaches, along with insights into their strengths and limitations in this environment.